Kingdom of Saudi Arabia
Ministry of Higher Education
King Faisal University
College of Computer Sciences & Information Technology

Hospital Database

Database Concepts and Design Course Project

Year: 2024 Semester: II

No.	Student Name	Student ID
S1	Rana Salah Althafar	222439076
S2	Amal Marshad M Salman	222421770
S3	Atekah Hussain Aljafar	222427138
S4	Hanan Abdullah Alsafran	222407860
S5	Maryam Ahmed Alshabib	223050271

Instructor: Dr. Bashayer Fareed Altulaihey

Table of Contents

Table of Contents 2
1. Introduction (case description) 3
2. System Analysis 4
A. List of users 4
B. List of main functions (at least three) 4
C. List of main reports (at least three) 4
3. Database Conceptual Design (ERD) 5
4. Logical Design (Relational Schema) 6
5. Physical Design 7
6. Populate Database 10
7. Queries 12
8. Application Development 14

1. Introduction

This database system is designed to manage patient information, treatment details, and staff assignments within a hospital emergency room (ER) environment. The system stores data in several tables, each representing a specific aspect of the ER operation:

Doctors: This table stores information about doctors working in the ER, including their unique ID, separate first and last name fields (Doctor_First_Name and Doctor_Last_Name), salary, and specialty (e.g., emergency medicine, pediatrics).

Departments: This table identifies different departments within the hospital (e.g., emergency room, radiology). Each department has a unique ID and a descriptive name.

Patients: This table stores patient information such as unique ID, separate first and last name fields (Patient_First_Name and Patient_Last_Name), age, sex, and address.
The "address" attribute can be considered a multivalued attribute. To manage this effectively, a separate table for addresses linked to the Patients table using a foreign key relationship is recommended. This allows for storing multiple addresses (e.g., home, work, mailing) with additional details like address type and dates.

Consults: This table establishes a many-to-many relationship between doctors and patients. It records details about each treatment session, including a unique ID, the doctor involved (linked to the Doctors table by doctor ID), the patient involved (linked to the Patients table by patient ID), date, and time of the session.

Room: This table identifies individual rooms within the ER, each with a unique room number and a type designation (e.g., examination room, treatment room).
Nurses: This table stores information about nurses working in the ER, including their unique ID, separate first and last name fields (Nurse_First_Name and Nurse_Last_Name), shift hours, age, salary, and **nurse's address (Nurse_Address)**.

Governs: This table establishes a many-to-many relationship between nurses and rooms. It includes a unique ID for each assignment, linking a nurse (by nurse ID) to a specific room (by room number).

2. System Analysis (We made some changes in this page)

A. List of users

Patient
Doctor
Nurse

B. List of main functions

1-Patients can book an appointment with a doctor.

2-Patients can view their own room number.

3-The nurse can view the room number for which he or she is responsible.

4-The doctor can view all the patients for whom he is responsible.

5-Doctor, Nurse and Patients can update their personal information in the system.

6- Nurse can update patient room number

7-Doctor and nurse can view their salary record.

C. List of main reports

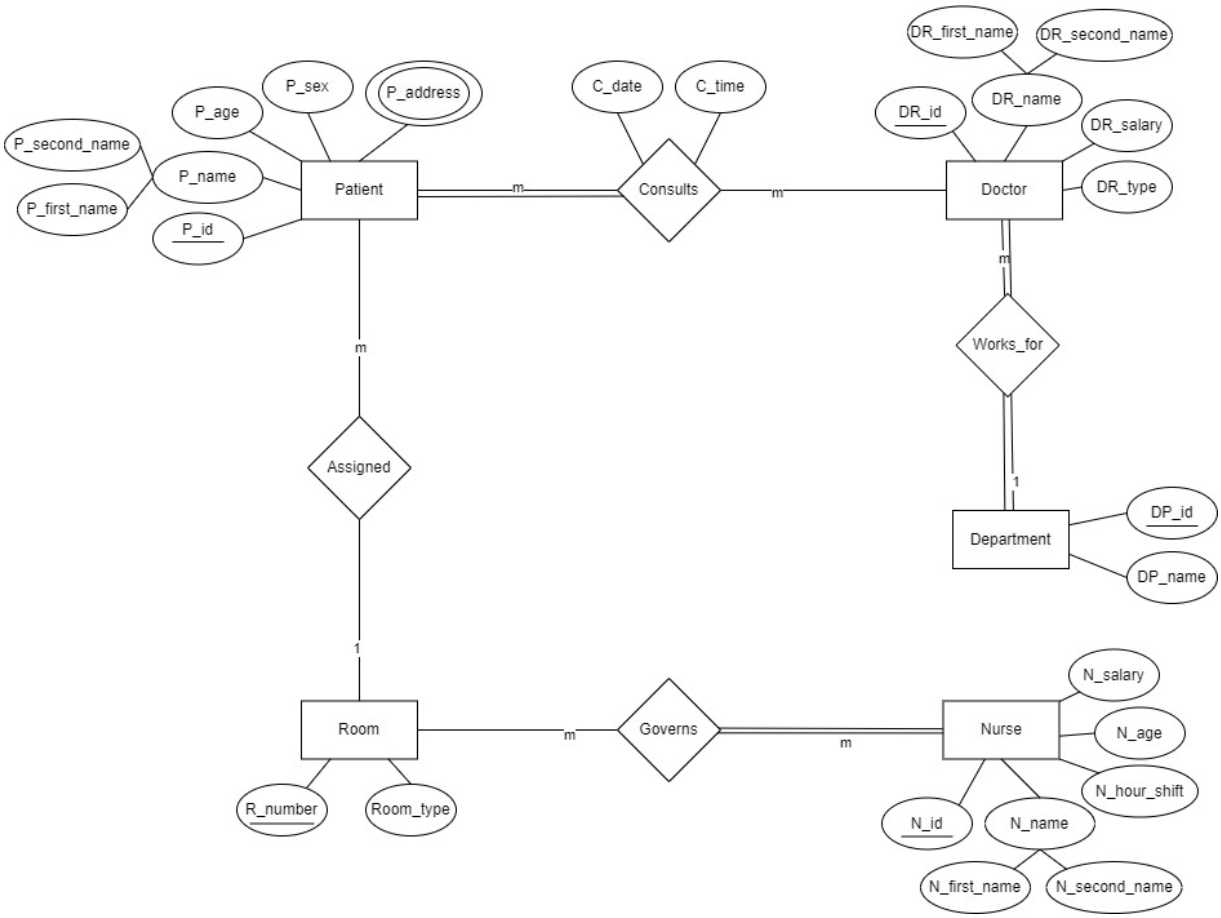
-Doctor can generate a list of expected patients.

-Patient can generate a list of all doctors.

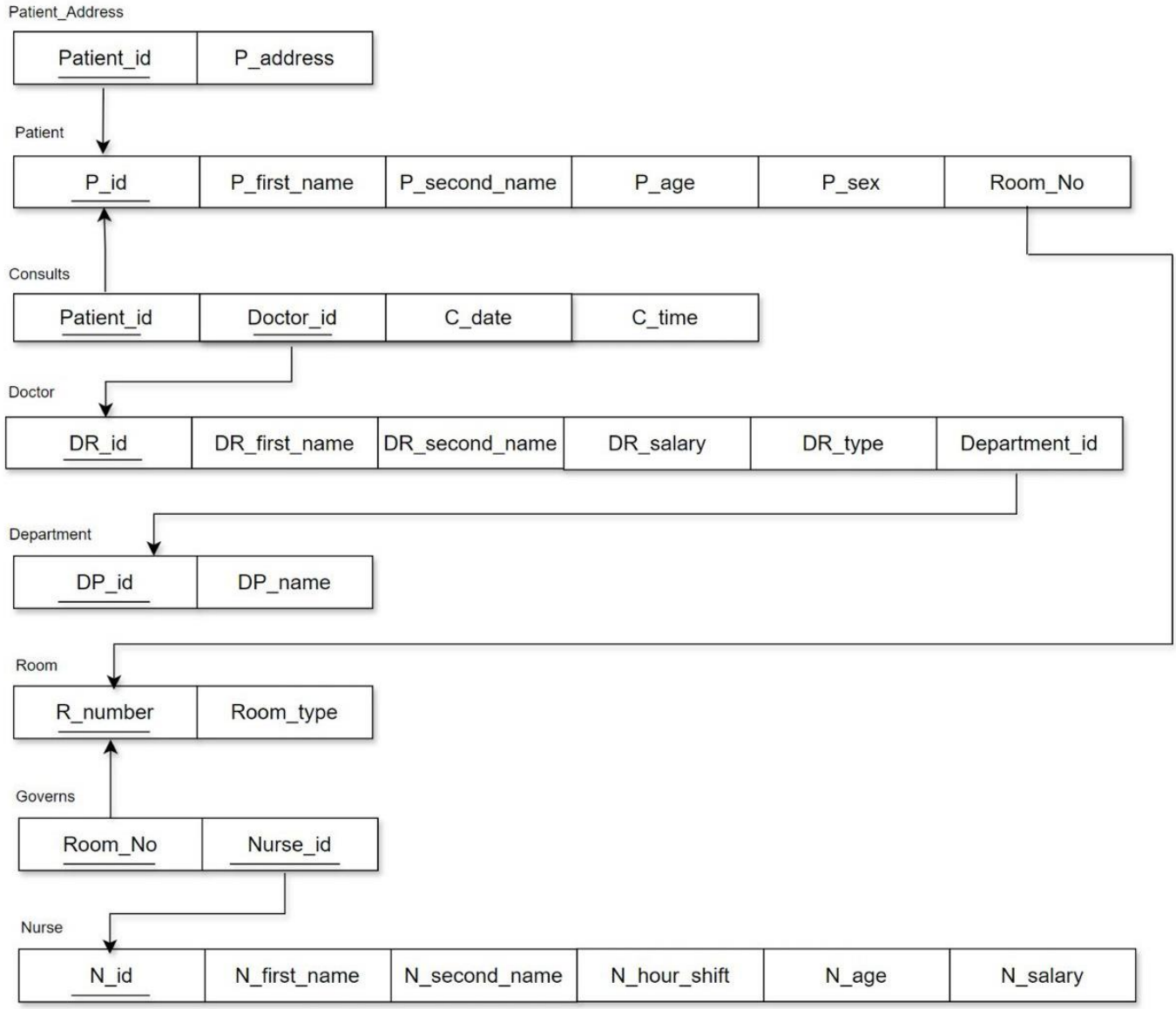
-Nurse can generate a list of all rooms.

-Patient can generate a list of his/her upcoming appointments.

3. Database Conceptual Design (ERD)



4. Logical Design (Relational Schema)



Physical Design

```sql
create table patient_address(
    patient_id int NOT NULL,
    patient_address varchar2(30),

    constraint patient_address_patient_id_pk primary key (patient_id)
)
```

Results | Explain | Describe | Saved SQL | History

Table created.

```sql
create table patient(
    patient_id int NOT NULL,
    patient_first_name varchar2(30) NOT NULL,
    patient_second_name varchar2(30) NOT NULL,
    patient_age int,
    patient_sex varchar2(30) CHECK( patient_sex in ('F', 'M')),
    room_number int,

    constraint patient_patient_id_pk primary key (patient_id),
    constraint patient_room_number_fk foreign key (room_number) references room(room_number) ON DELETE CASCADE
)
```

Results | Explain | Describe | Saved SQL | History

Table created.

```sql
create table doctor(
    doctor_id int NOT NULL,
    doctor_first_name varchar2(30) NOT NULL,
    doctor_second_name varchar2(30) NOT NULL,
    doctor_salary int not null,
    doctor_type varchar2(30),
    department_id int,

    constraint doctor_doctor_id_pk primary key(doctor_id),
    constraint doctor_department_id foreign key(department_id) references department(department_id) ON DELETE CASCADE
)
```

Results | Explain | Describe | Saved SQL | History

Table created.

```sql
create table department(
    department_id int NOT NULL,
    department_name varchar2(30) NOT NULL,

    constraint department_department_id_pk primary key(department_id)
)
```

Results | Explain | Describe | Saved SQL | History

Table created.

```sql
create table room(
    room_number int NOT NULL,
    room_type varchar2(30),

    constraint room_room_number_pk primary key(room_number)
)
```

Results | Explain | Describe | Saved SQL | History

Table created.

```sql
create table consults(
    patient_id int NOT NULL,
    doctor_id int NOT NULL,
    consults_date varchar2(10) NOT NULL ,
    consults_time varchar2(5) NOT NULL ,

    constraint consults_patient_id_pk primary key (patient_id, doctor_id),
    constraint consults_patient_id_fk foreign key (patient_id) references patient(patient_id) ON DELETE CASCADE,
    constraint consults_doctor_id_fk foreign key (doctor_id)references doctor(doctor_id) ON DELETE CASCADE
)
```

Results | Explain | Describe | Saved SQL | History

Table created.

```sql
create table nurse(
    nurse_id int NOT NULL,
    nurse_first_name varchar2(30) NOT NULL,
    nurse_second_name varchar2(30) NOT NULL,
    nurse_hour_shift int,
    nurse_age int ,
    nurse_salary int not null,

    constraint nurse_nurse_id_pk primary key(nurse_id)

)
```

Results | Explain | Describe | Saved SQL | History

Table created.

```sql
create table governs(
    room_number int,
    nurse_id int,

    constraint governs_room_number_pk primary key(room_number,nurse_id),
    constraint governs_room_number_fk foreign key(room_number) references room(room_number) ON DELETE CASCADE ,
    constraint governs_nurse_id_fk foreign key(nurse_id) references nurse(nurse_id) ON DELETE CASCADE
)
```

Results | Explain | Describe | Saved SQL | History

Table created.

Populate Database

CONSULTS

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

PATIENT_ID	DOCTOR_ID	CONSULTS_DATE	CONSULTS_TIME
14325	3003	2024-05-03	13:15
67890	4004	2024-05-05	11:00
3010	2002	2024-05-08	12:45
4002	1001	2024-05-01	09:30

DEPARTMENT

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

DEPARTMENT_ID	DEPARTMENT_NAME
201	Cardiology
401	Physical Therapy
301	Surgery
101	Emergency Medicine

DOCTOR

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

DOCTOR_ID	DOCTOR_FIRST_NAME	DOCTOR_SECOND_NAME	DOCTOR_SALARY	DOCTOR_TYPE	DEPARTMENT_ID
2002	Ethan	Martinez	250000	Physical Therapy Assistant	401
4004	Benjamin	Adams	220000	Emergency Medicine Depart...	101
1001	Olivia	Johnson	180000	Cardiologist	201
3003	Sophia	Lee	300000	Neurosurgeon	301

GOVERNS

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

ROOM_NUMBER	NURSE_ID
301	1007
402	1009
409	1010
502	1008

NURSE

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

NURSE_ID	NURSE_FIRST_NAME	NURSE_SECOND_NAME	NURSE_HOUR_SHIFT	NURSE_AGE	NURSE_SALARY
1010	Mia	Lewis	8	28	5200
1007	Noah	Taylor	8	33	5100
1009	Liam	Clark	10	26	4800
1008	Isabella	Harris	12	30	5900

PATIENT

Columns | **Data** | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

PATIENT_ID	PATIENT_FIRST_NAME	PATIENT_SECOND_NAME	PATIENT_AGE	PATIENT_SEX	ROOM_NUMBER
4002	Jack	Wilson	45	M	
14325	Emily	Johnson	42	F	301
67890	Michael	Anderson	27	M	
3010	Sarah	Thompson	52	F	502

PATIENT_ADDRESS

Columns | **Data** | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

PATIENT_ID	PATIENT_ADDRESS
14325	Via Roma 10, Milan, Italy
4002	Via Roma 5, Naples, Italy
3010	Via Cavour 12, Florence, Italy
67890	Via Venezia 15, Naples, Italy

ROOM

Columns | **Data** | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

ROOM_NUMBER	ROOM_TYPE
409	Isolation Room
301	Special Care Unit
402	Maternity Suite
502	Palliative Care Room

Queries

```sql
SELECT DOCTOR_FIRST_NAME ||' '|| DOCTOR_SECOND_NAME AS DOCTOR_FULL_NAME, DEPARTMENT_NAME
FROM doctor
join department
on doctor.DEPARTMENT_ID = department.DEPARTMENT_ID;
```

Results | Explain | Describe | Saved SQL | History

DOCTOR_FULL_NAME	DEPARTMENT_NAME
Ethan Martinez	Physical Therapy
Benjamin Adams	Emergency Medicine
Olivia Johnson	Cardiology
Sophia Lee	Surgery

4 rows returned in 0.02 seconds Download

```sql
SELECT PATIENT_FIRST_NAME ||' '||PATIENT_SECOND_NAME AS PATIENT_FULL_NAME,CONSULTS_DATE,CONSULTS_TIME,DOCTOR_ID
FROM patient
right outer join consults
on patient.PATIENT_ID = consults.PATIENT_ID;
```

Results | Explain | Describe | Saved SQL | History

PATIENT_FULL_NAME	CONSULTS_DATE	CONSULTS_TIME	DOCTOR_ID
Emily Johnson	2024-05-03	13:15	3003
Michael Anderson	2024-05-05	11:00	4004
Sarah Thompson	2024-05-08	12:45	2002
Jack Wilson	2024-05-01	09:30	1001

4 rows returned in 0.01 seconds Download

```sql
SELECT * from doctor
where doctor_salary > 250000;
```

Results | Explain | Describe | Saved SQL | History

DOCTOR_ID	DOCTOR_FIRST_NAME	DOCTOR_SECOND_NAME	DOCTOR_SALARY	DOCTOR_TYPE	DEPARTMENT_ID
3003	Sophia	Lee	300000	Neurosurgeon	301

1 rows returned in 0.00 seconds Download

```sql
SELECT PATIENT_FIRST_NAME ,room.ROOM_NUMBER,patient_id
FROM patient JOIN room
ON patient.ROOM_NUMBER = room.ROOM_NUMBER
WHERE room.ROOM_NUMBER is not null
ORDER BY patient_id DESC ;
```

Results | Explain | Describe | Saved SQL | History

PATIENT_FIRST_NAME	ROOM_NUMBER	PATIENT_ID
Emily	301	14325
Sarah	502	3010

2 rows returned in 0.01 seconds Download

```sql
SELECT DISTINCT NURSE_HOUR_SHIFT,NURSE_FIRST_NAME AS NURSE_NAME , NURSE_SALARY
FROM nurse
WHERE NURSE_AGE < 31;
```

Results | Explain | Describe | Saved SQL | History

NURSE_HOUR_SHIFT	NURSE_NAME	NURSE_SALARY
8	Mia	5200
12	Isabella	5900
10	Liam	4800

3 rows returned in 0.01 seconds Download

```sql
SELECT patient_address,patient_first_name AS patient_name , patient_age
FROM PATIENT join PATIENT_ADDRESS
ON PATIENT.patient_id = PATIENT_ADDRESS.patient_id
WHERE patient_address LIKE '%N%'
ORDER BY patient_name;
```

Results | Explain | Describe | Saved SQL | History

PATIENT_ADDRESS	PATIENT_NAME	PATIENT_AGE
Via Roma 5, Naples, Italy	Jack	45
Via Venezia 15, Naples, Italy	Michael	27

2 rows returned in 0.01 seconds Download

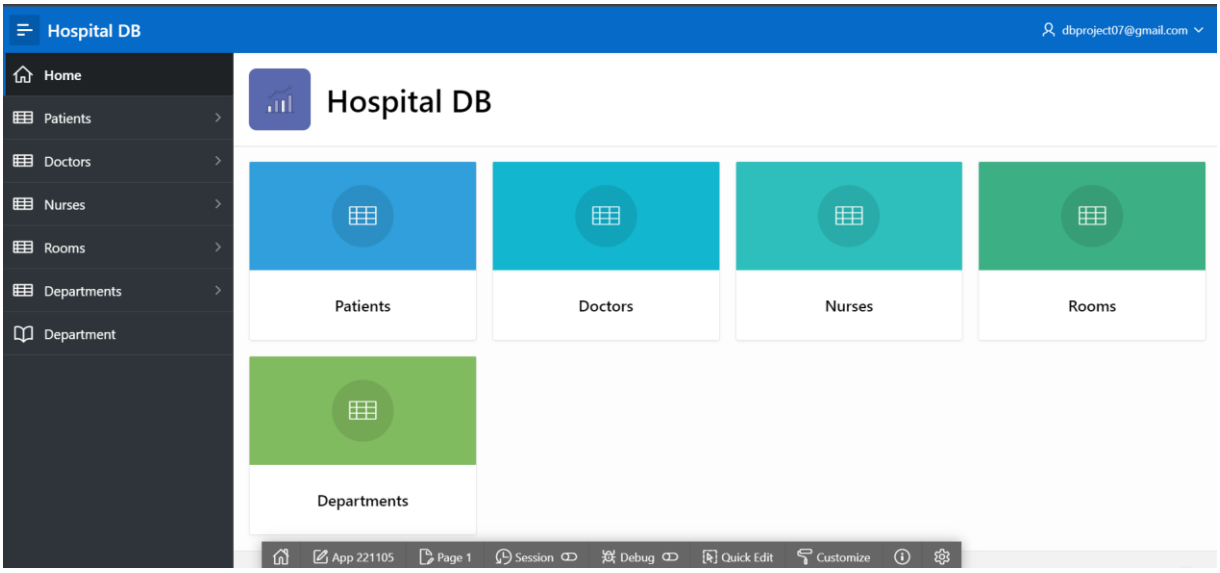
```sql
SELECT COUNT(nurse_id) AS number_of_nurse ,nurse_hour_shift
from GOVERNS NATURAL JOIN NURSE
GROUP BY nurse_hour_shift;
```

Results | Explain | Describe | Saved SQL | History

NUMBER_OF_NURSE	NURSE_HOUR_SHIFT
2	8
1	12
1	10

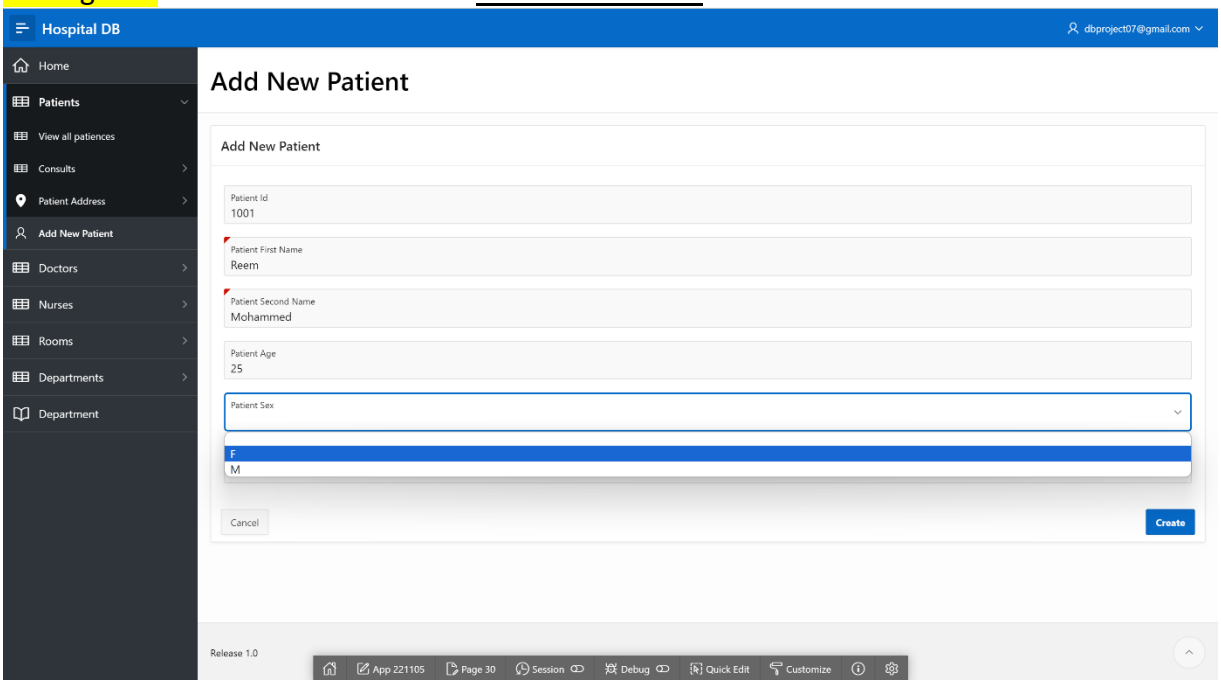
Application Development

-Home page

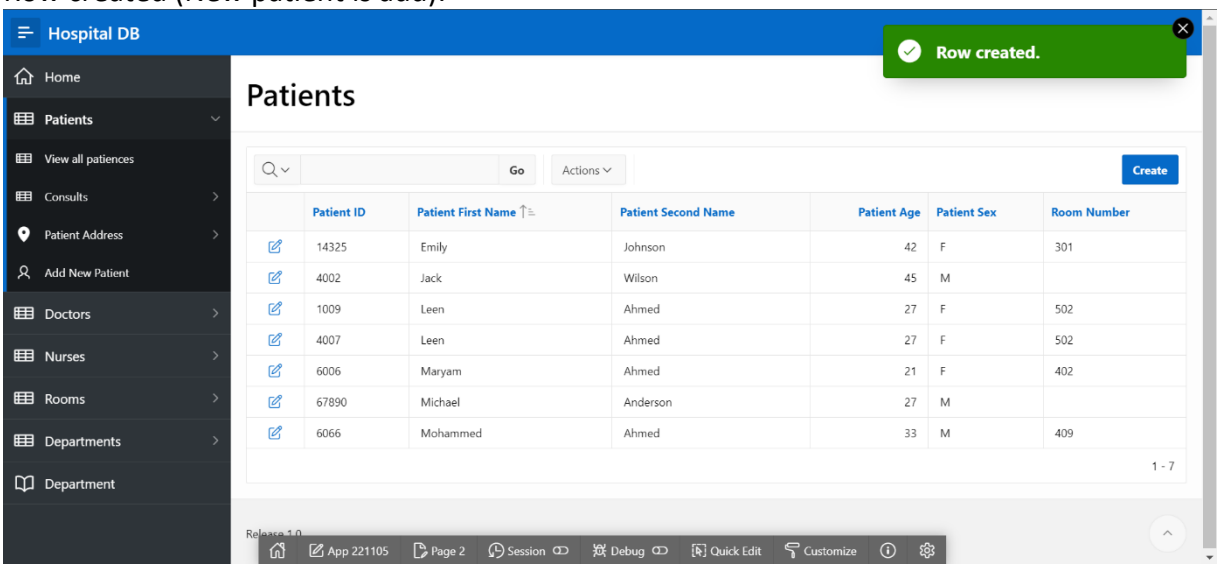


- Add new data to tables using forms. Apply validation on each field (e.g. if Primary key is duplicate, It should display an error message)

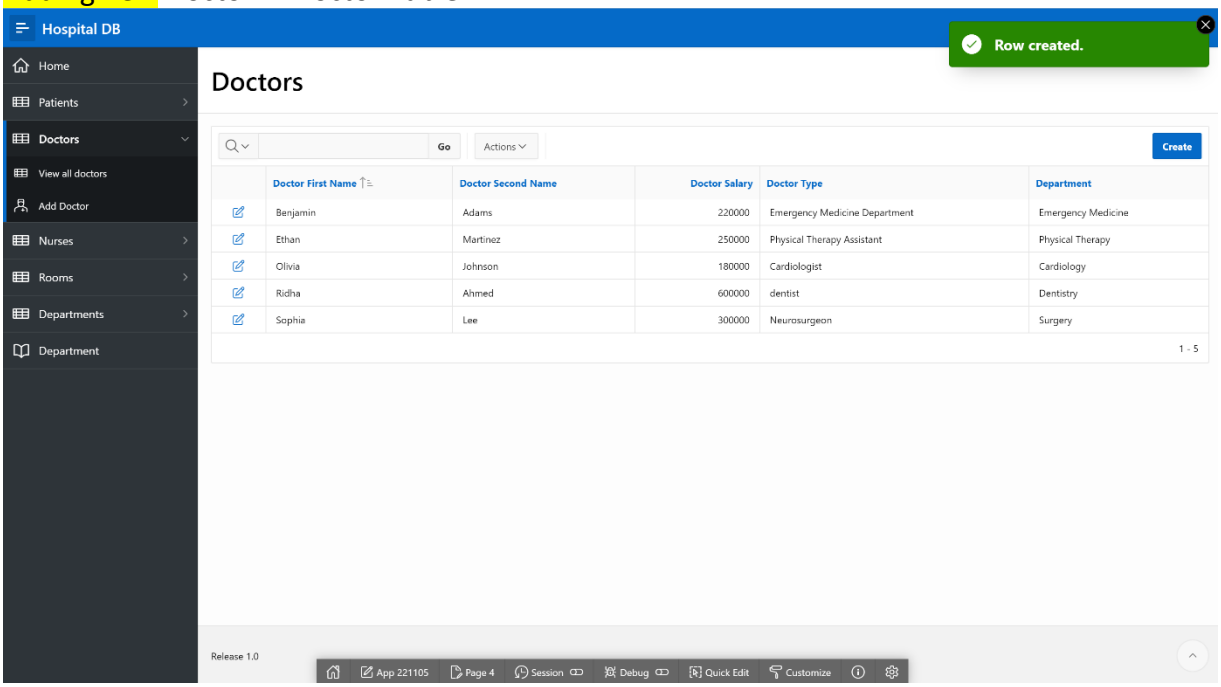
Adding new Patient in Patient Table Use Selection List:



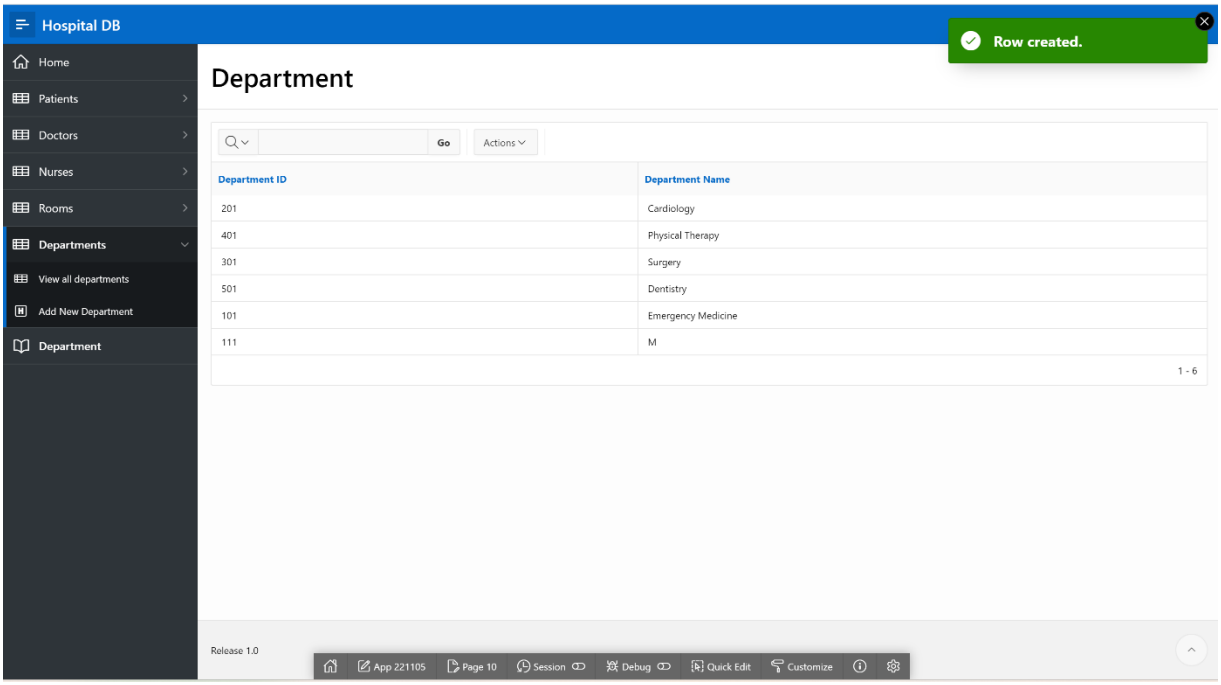
Row created (New patient is add):



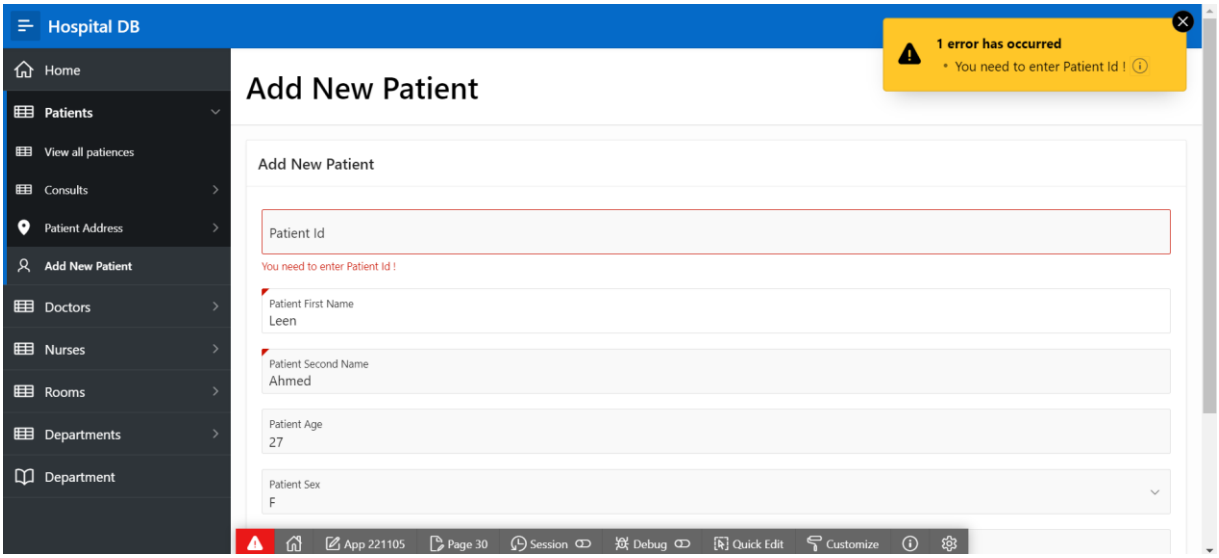
Adding new Doctor in Doctor Table:



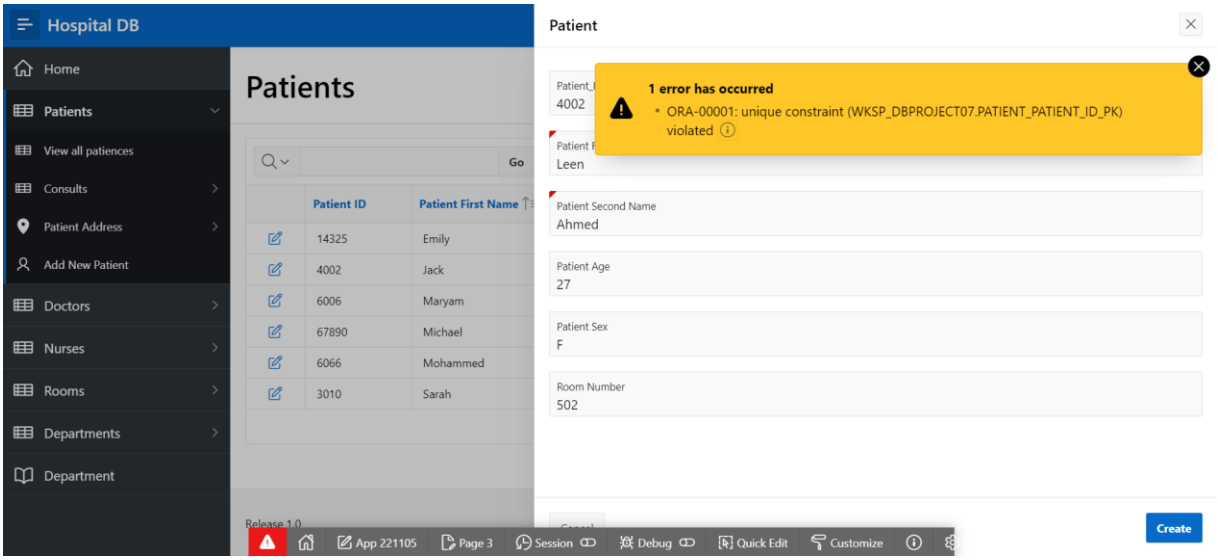
Adding new Department in Department Table:



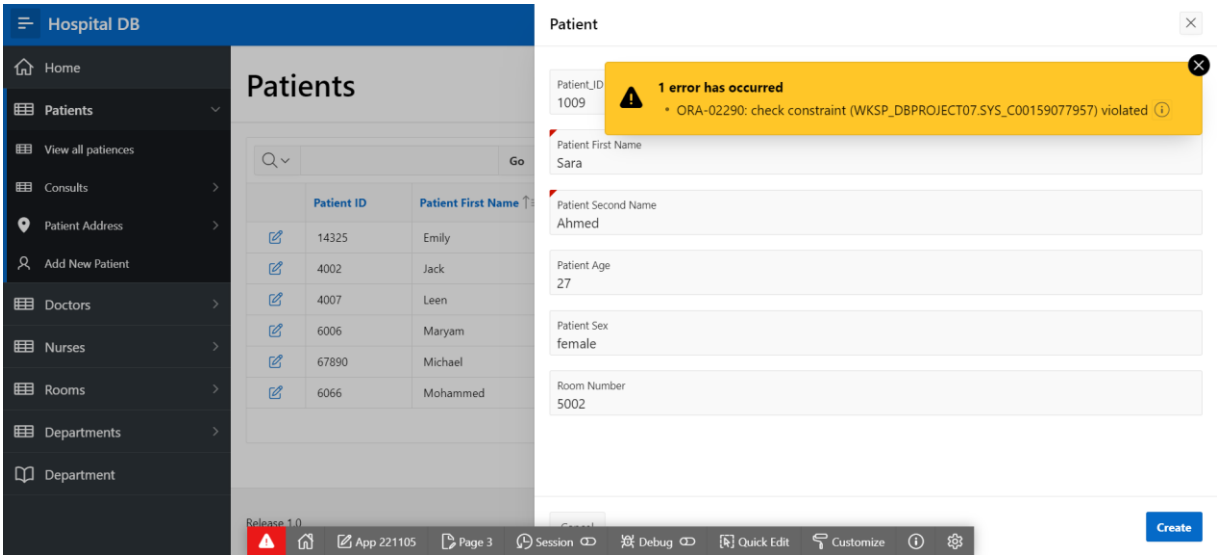
Display error message when Primary key is empty:



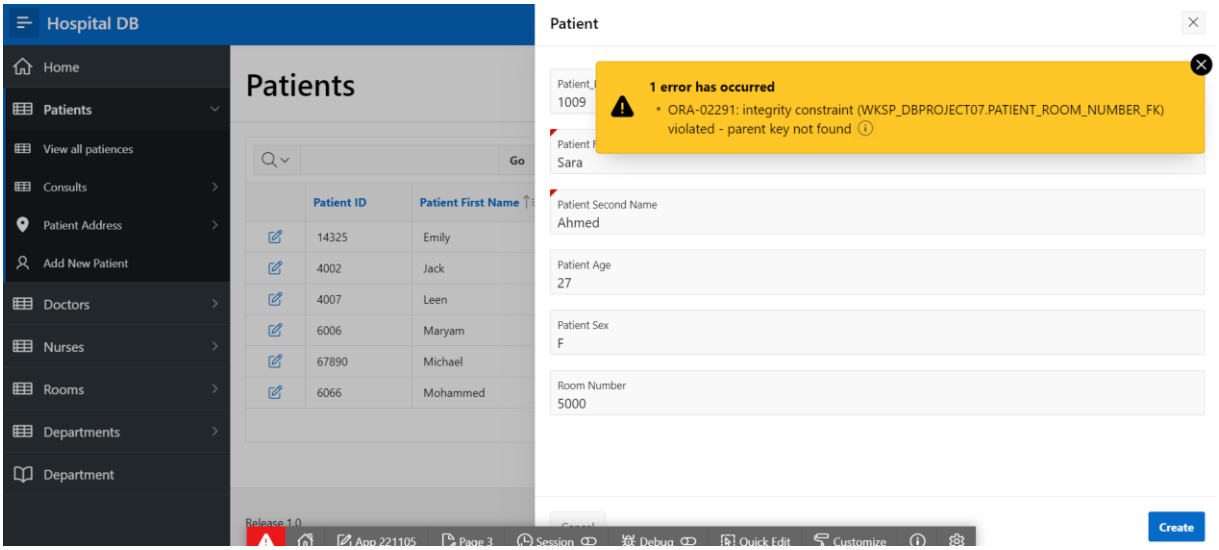
Display error message when Primary key is duplicate:



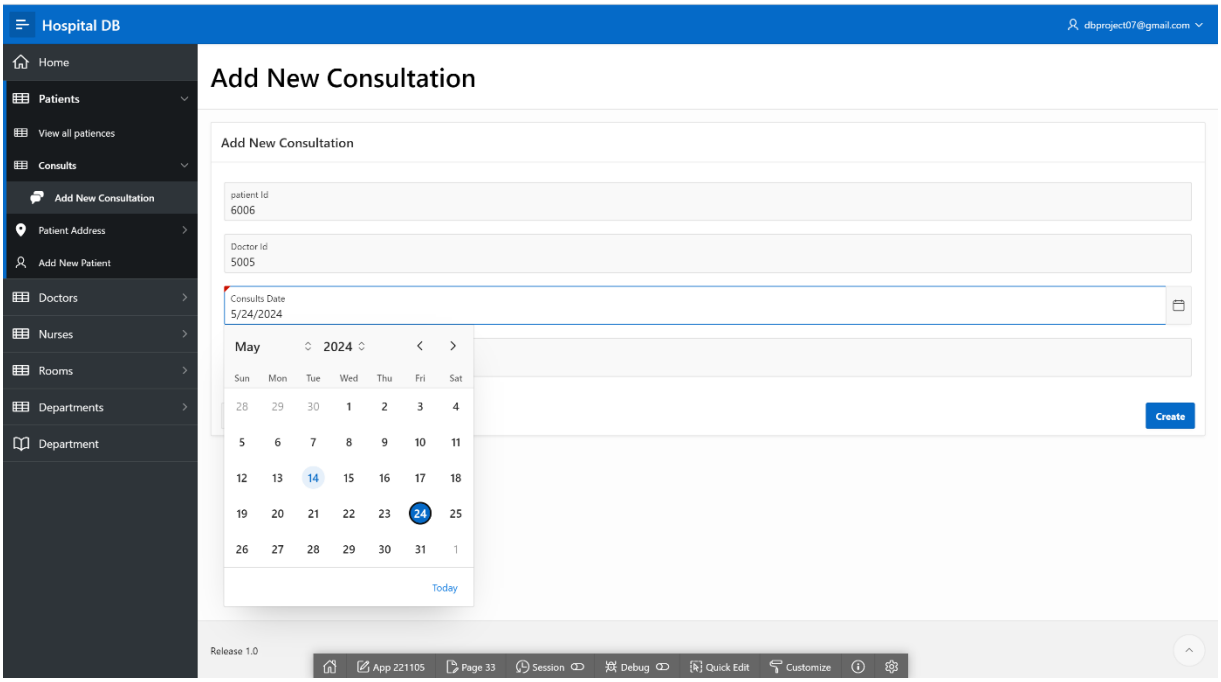
Check constrains is violated:

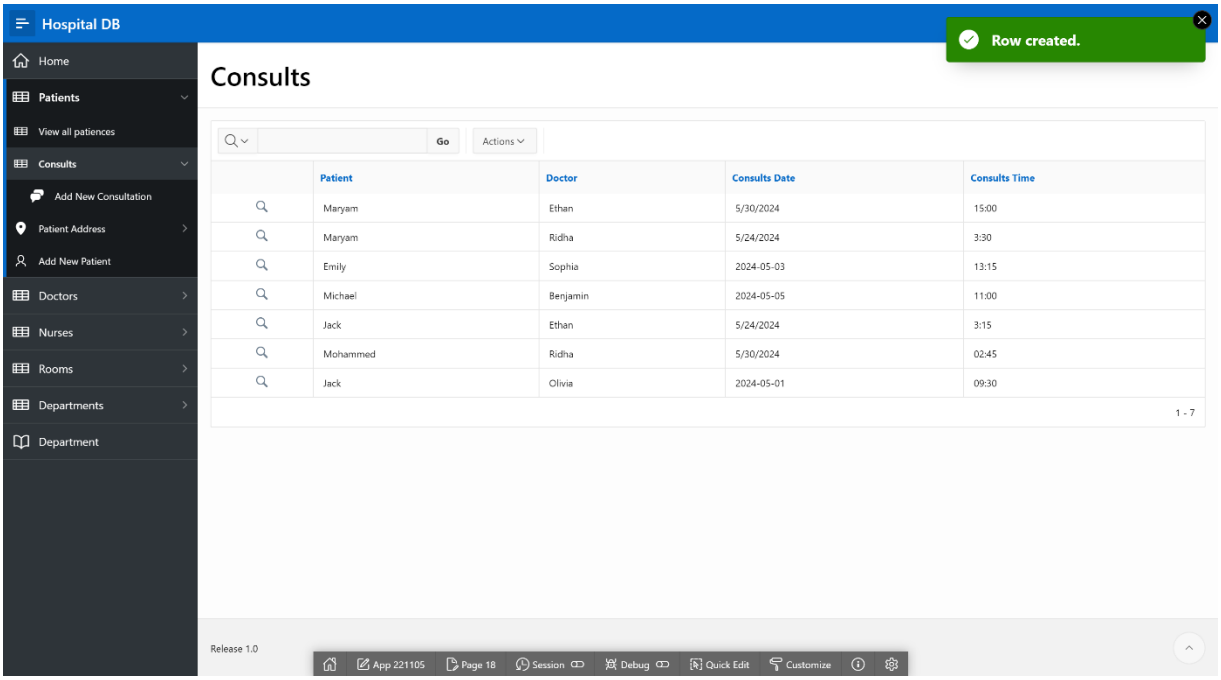


Display error message when Foreign Key is not fount:



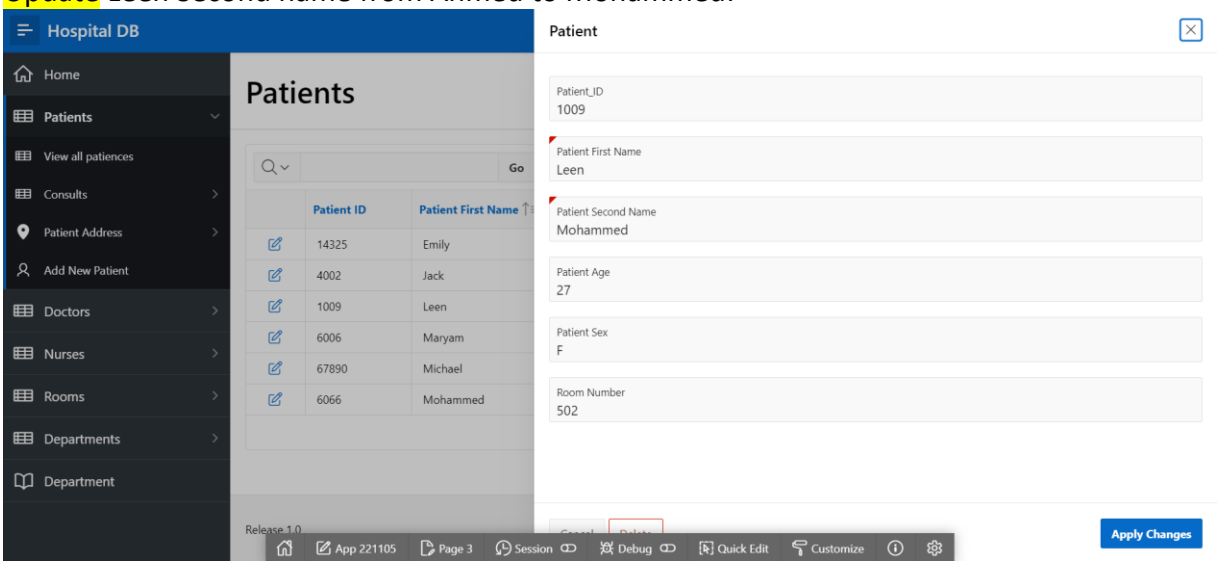
Add new consultation in consults table using Data Type:

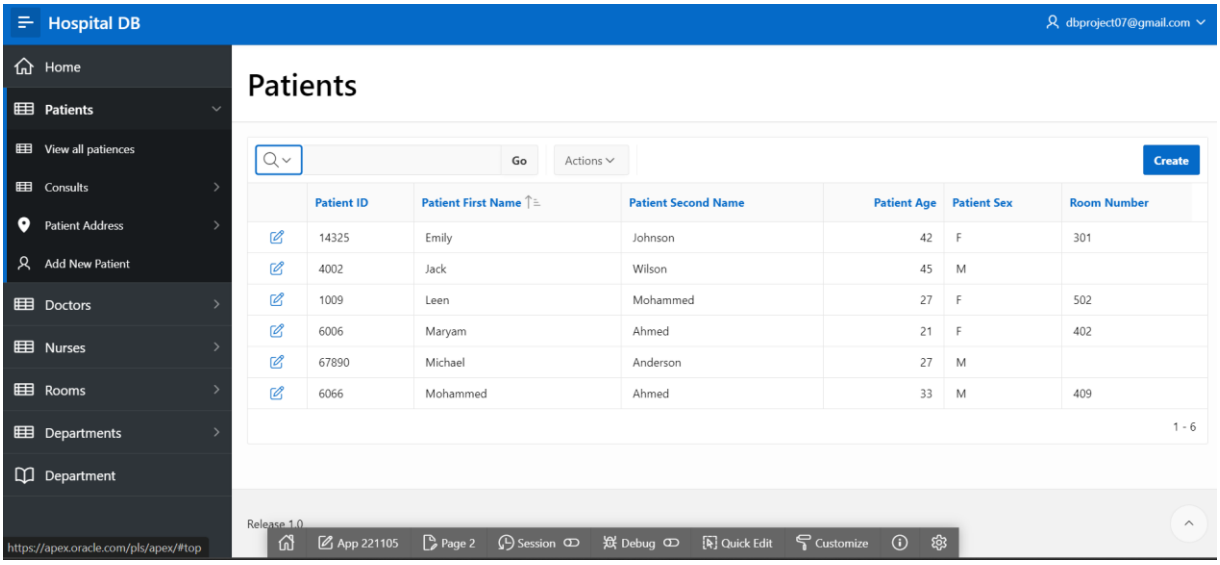



Hospital DB
Row created.
Home
Patients
View all patiences
Consults
Add New Consultation
Patient Address
Add New Patient
Doctors
Nurses
Rooms
Departments
Department
Consults
Go
Actions
Patient
Doctor
Consults Date
Consults Time
Maryam Ethan 5/30/2024 15:00
Maryam Ridha 5/24/2024 3:30
Emily Sophia 2024-05-03 13:15
Michael Benjamin 2024-05-05 11:00
Jack Ethan 5/24/2024 3:15
Mohammed Ridha 5/30/2024 02:45
Jack Olivia 2024-05-01 09:30
1 - 7
Release 1.0
App 221105
Page 18
Session
Debug
Quick Edit
Customize

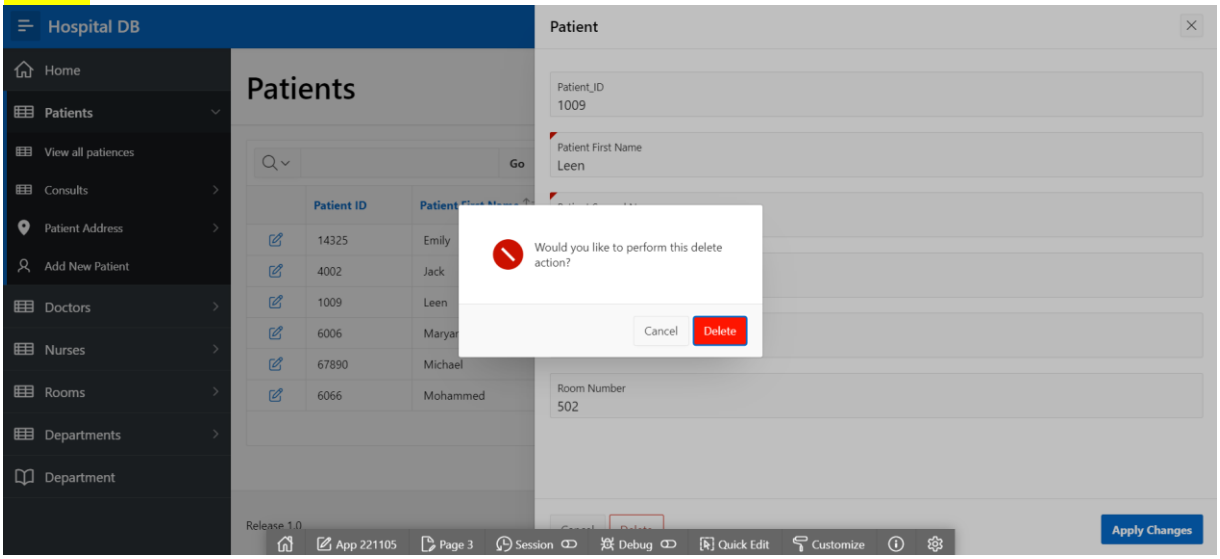
- Delete/ update record from each table

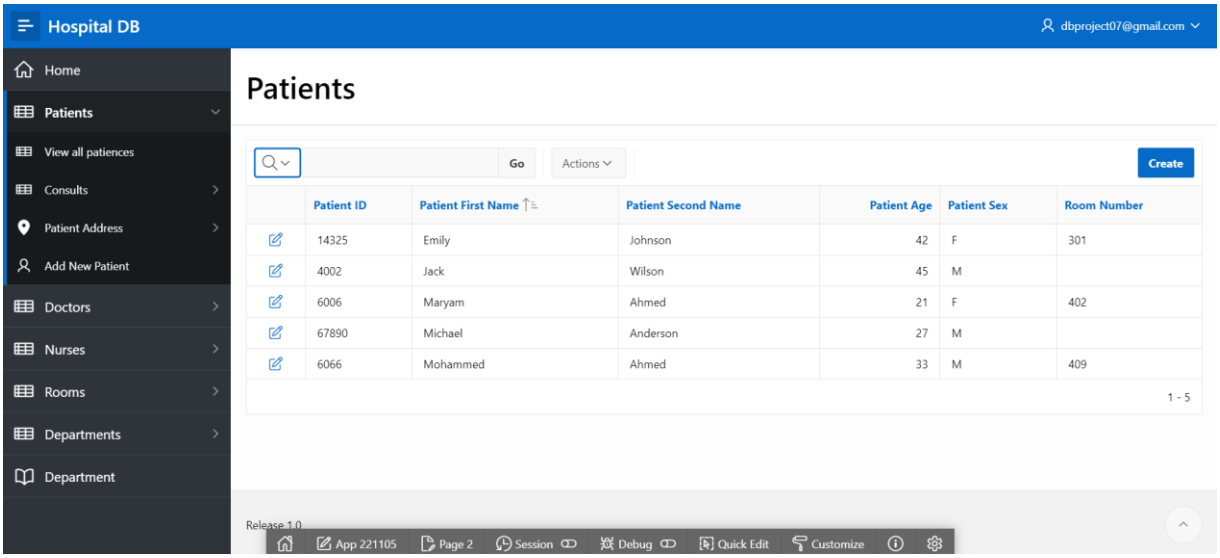
Update Leen Second name from Ahmed to Mohammed:





Delete Leen Mohammed:



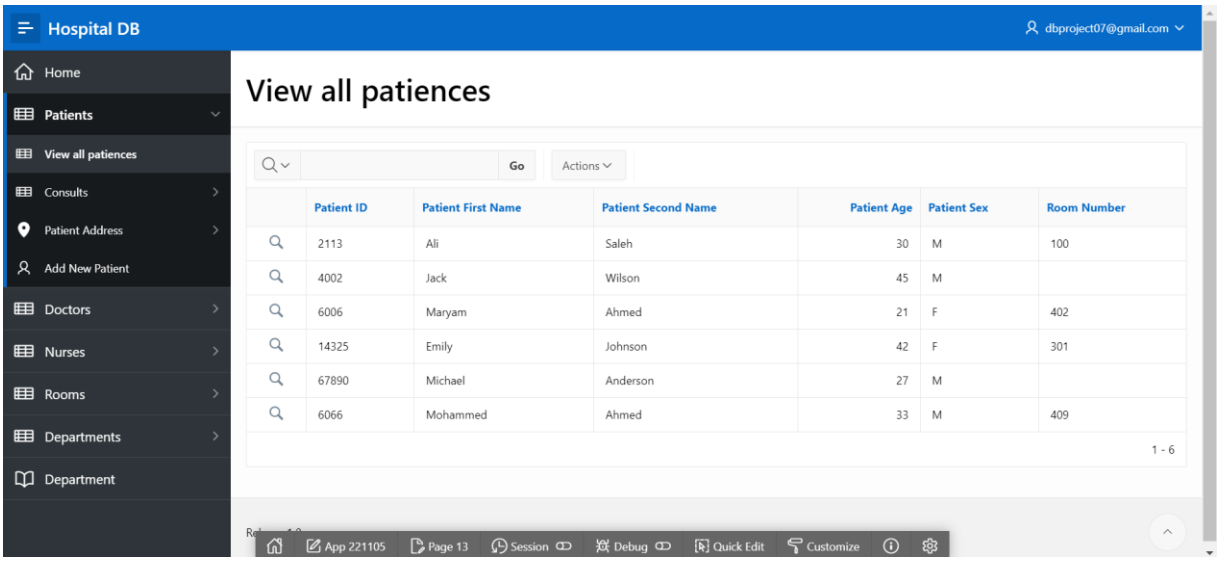


- If a record which refers to another table is deleted or updated; apply CASCADE Update and Cascade Delete (Primary foreign key relationship)

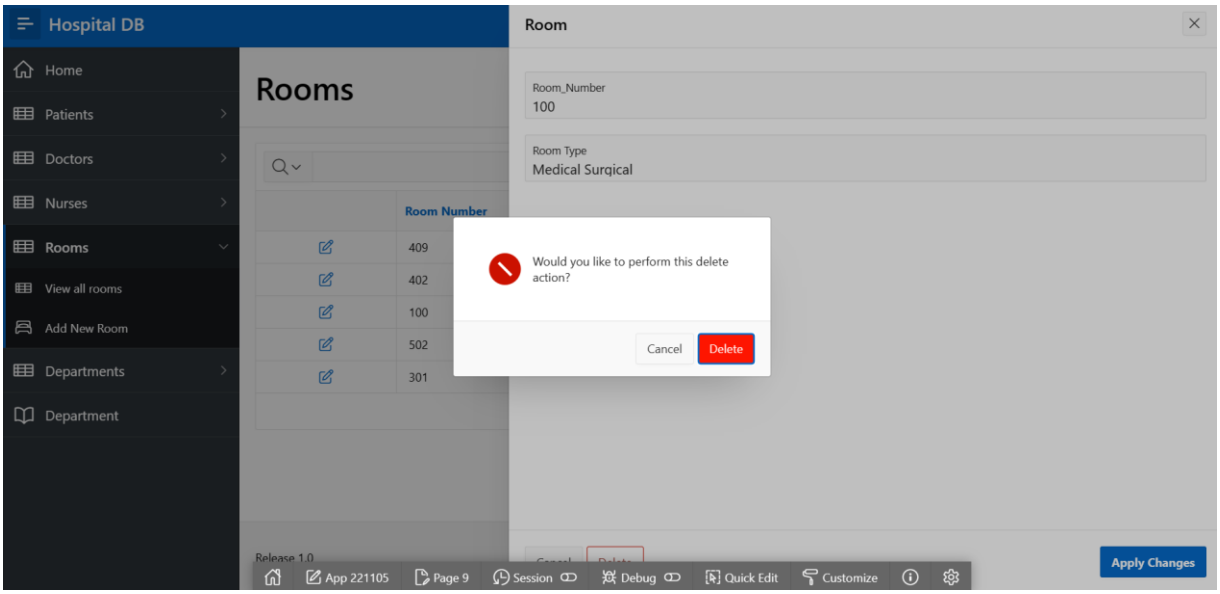
Applying CASCADE on column Room number in patient table

Columns	Delete Rule
ROOM_NUMBER	CASCADE

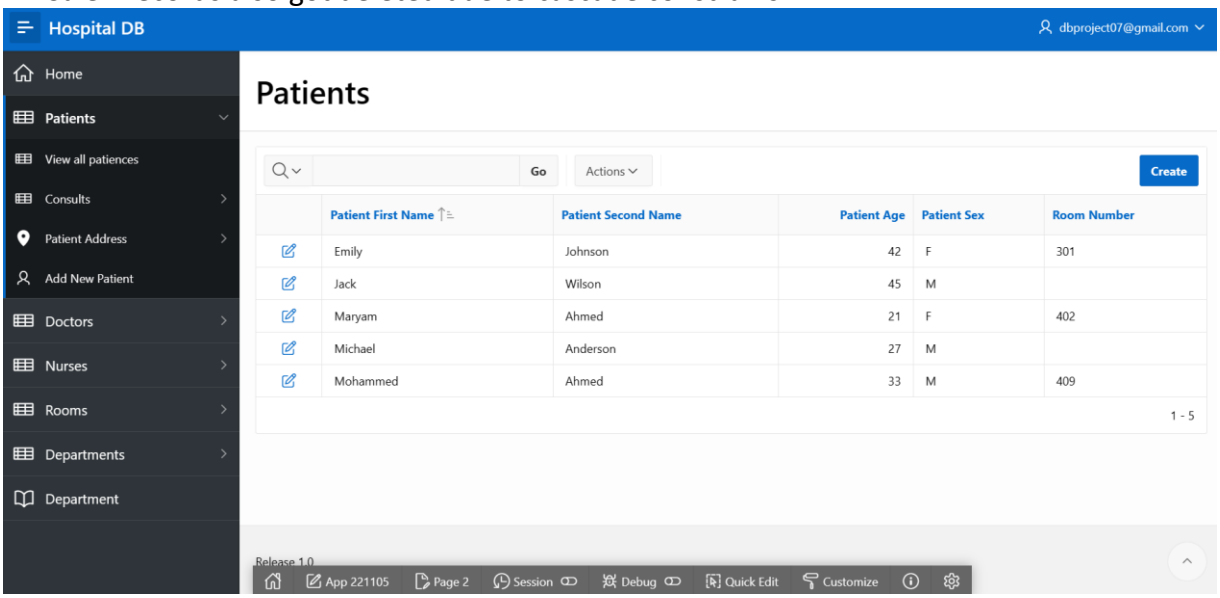
Ali Saleh is in room 100:



Delete Room number 100:



Ali Saleh records also got deleted due to cascade constrains:



For cascade update it did not work for us. We searched and tried to follow this approach:

```
-- Step 1: Drop existing foreign key
constraint (if it exists)
ALTER TABLE Orders
DROP CONSTRAINT FK_CustomerID;

-- Step 2: Add new foreign key
constraint with ON UPDATE CASCADE and ON
DELETE CASCADE
ALTER TABLE Orders
ADD CONSTRAINT FK_CustomerID
FOREIGN KEY (CustomerID)
REFERENCES Customers(CustomerID)
ON UPDATE CASCADE
ON DELETE CASCADE;
```

And this what it shows for us:

```
1  Alter table PATIENT
2  (
3      DROP CONSTRAINT PATIENT_ROOM_NUMBER_FK;
4  )
```

Results | Explain | Describe | Saved SQL | History

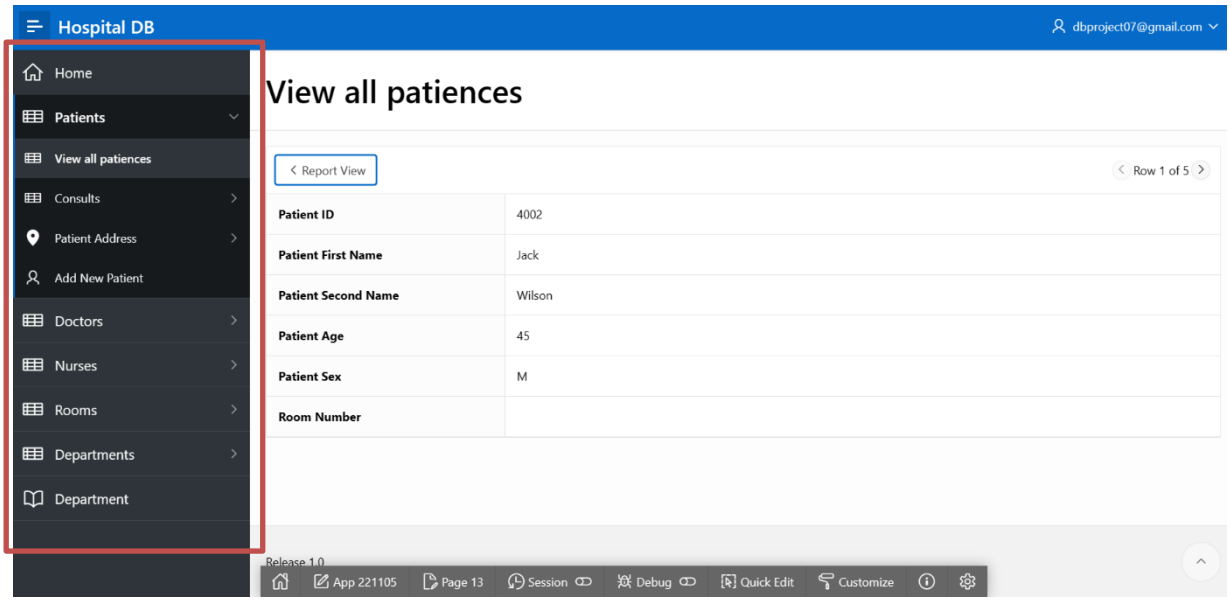
```
Error at line 2/1: ORA-01735: invalid ALTER TABLE option


1. Alter table PATIENT
2. (
3.     DROP CONSTRAINT PATIENT_ROOM_NUMBER_FK;
4. )
```

```
1  Alter table PATIENT
2  ADD CONSTRAINT PATIENT_ROOM_NUMBER_FK FOREIGN KEY (room_number) REFERENCE ROOM(room_number) ON DELETE CASCADE ON UPDATE CASCADE;
```

Results | Explain | Describe | Saved SQL | History

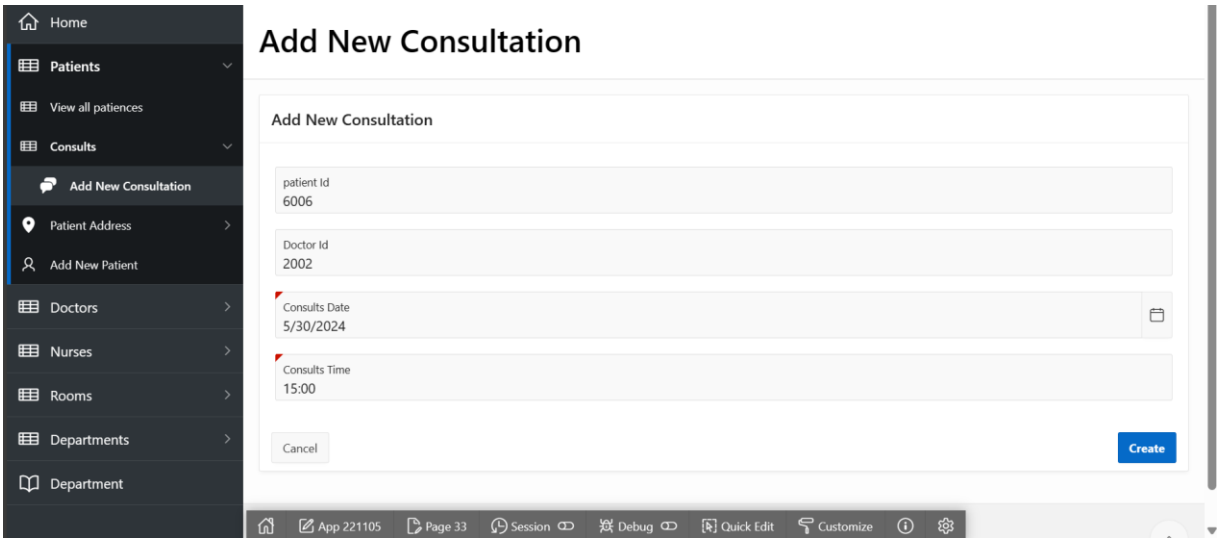
```
Error at line 2/65: ORA-00905: missing keyword


1. Alter table PATIENT
2. ADD CONSTRAINT PATIENT_ROOM_NUMBER_FK FOREIGN KEY (room_number) REFERENCE
ROOM(room_number) ON UPDATE CASCADE;
```

- Navigation between records (e.g. MoveFirst, MoveLast, Next, Previous etc..)

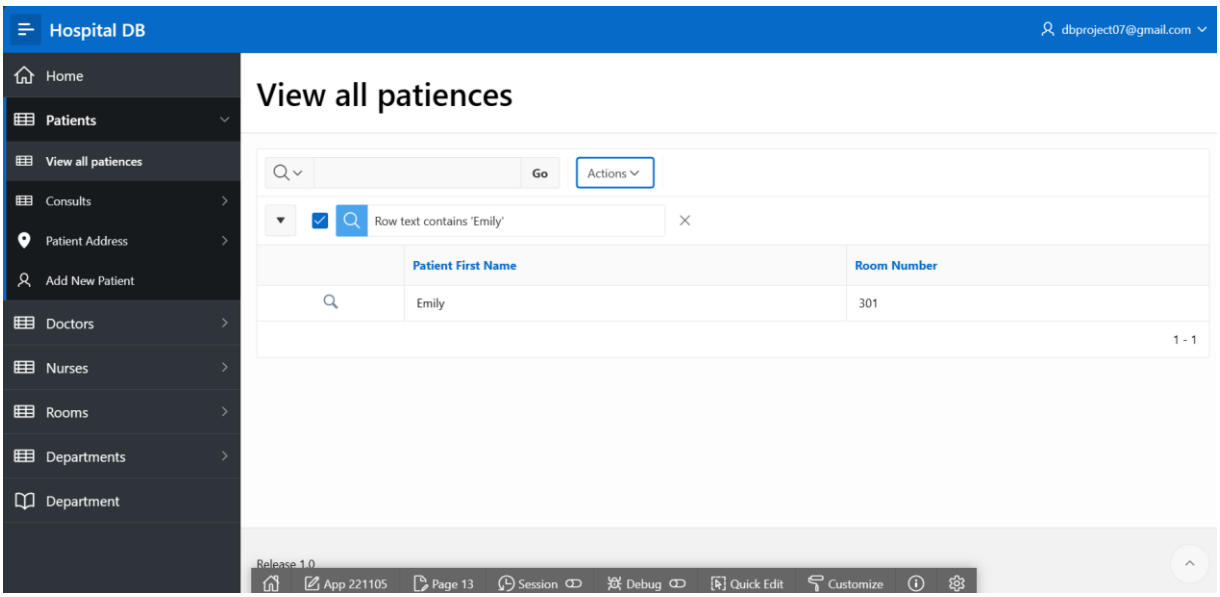


- Implement each business functions for different users listed under sect 2-B

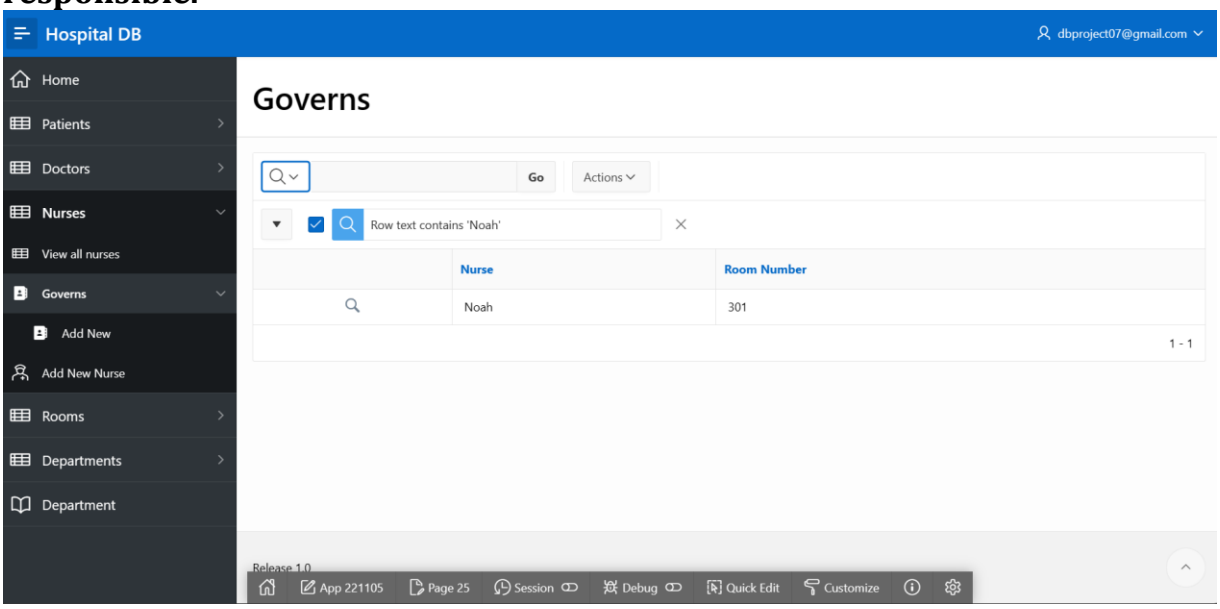
1-Patients can book an appointment with a doctor.



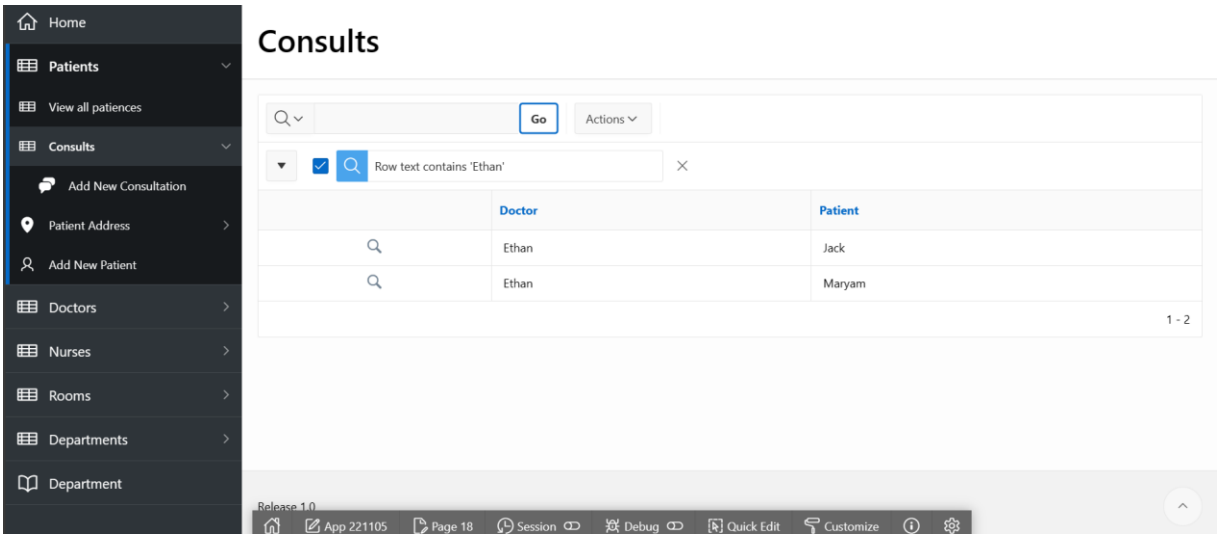
2-Patients can view their own room number.



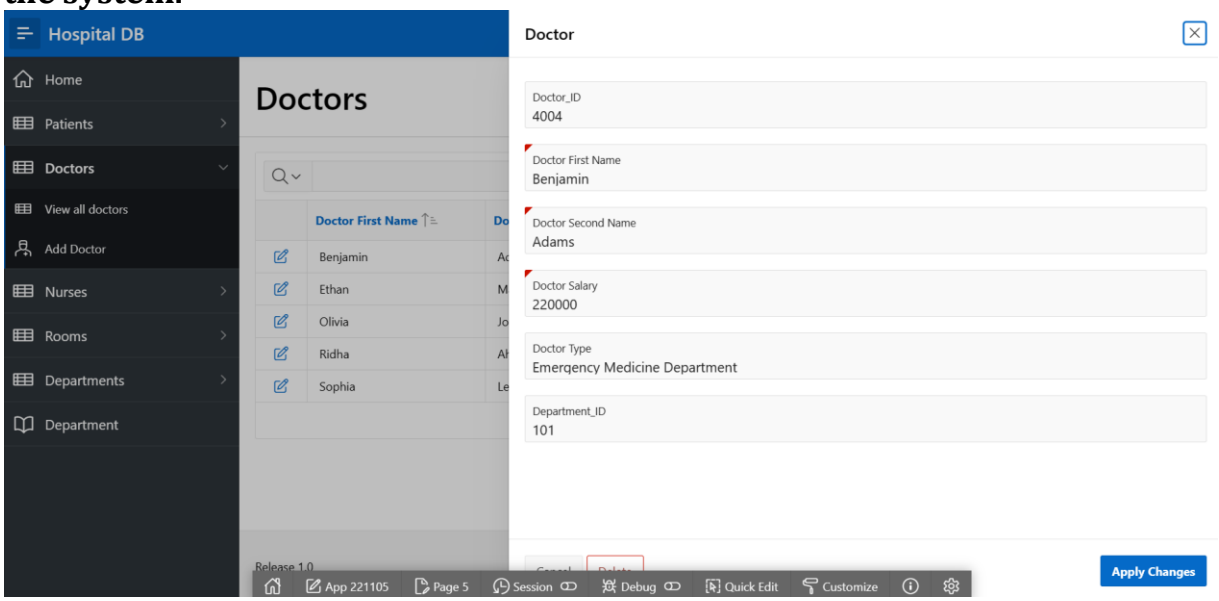
3-The nurse can view the room number for which he or she is responsible.



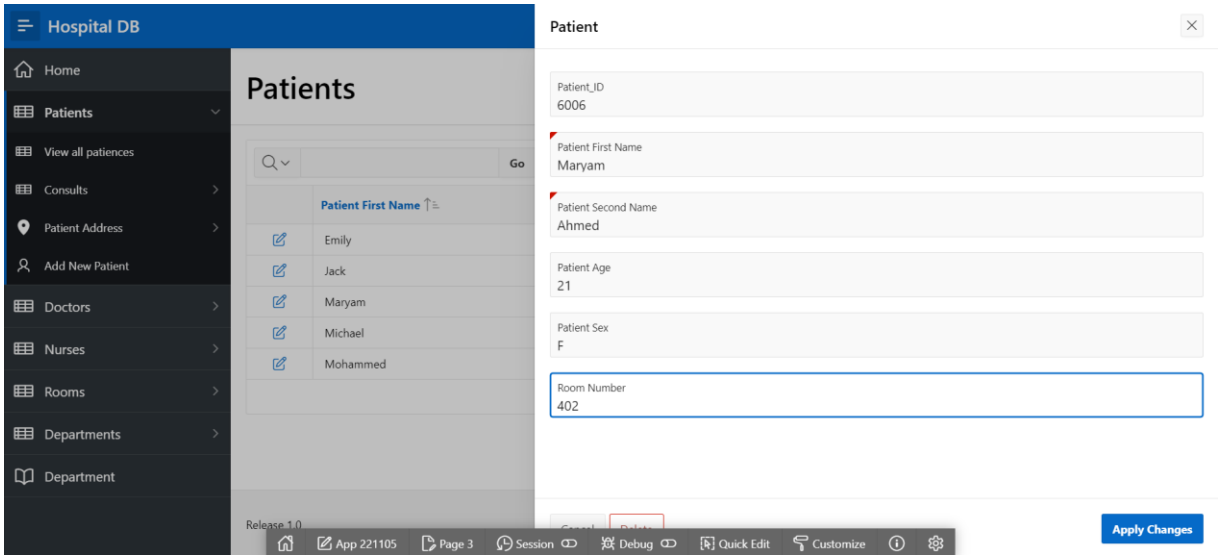
4-The doctor can view all the patients for whom he is responsible.



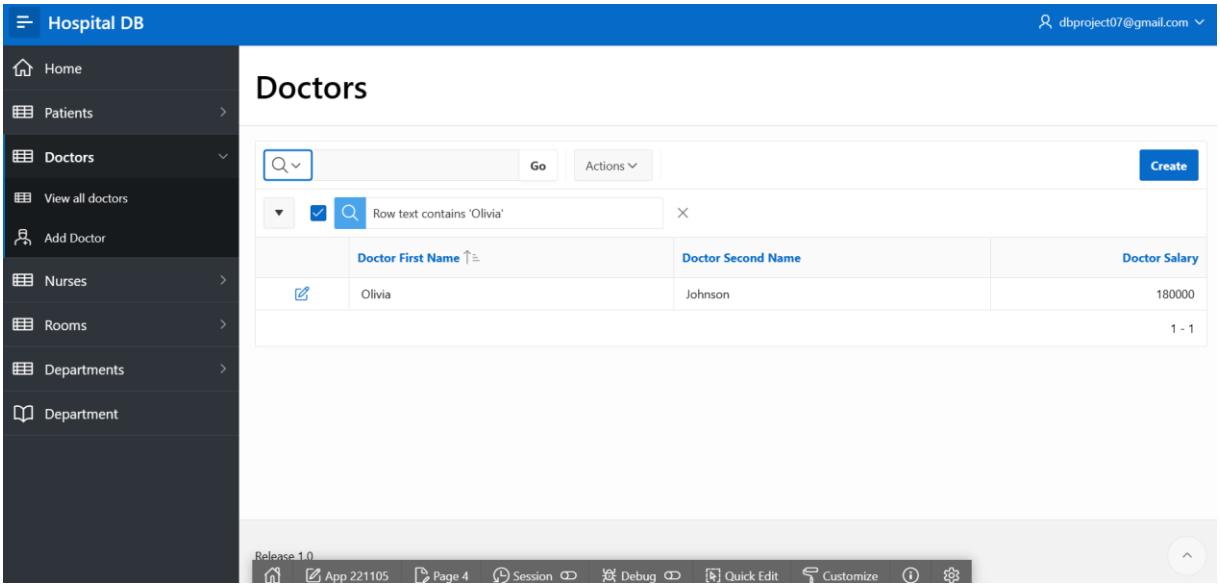
5-Doctor, Nurse and Patients can update their personal information in the system.

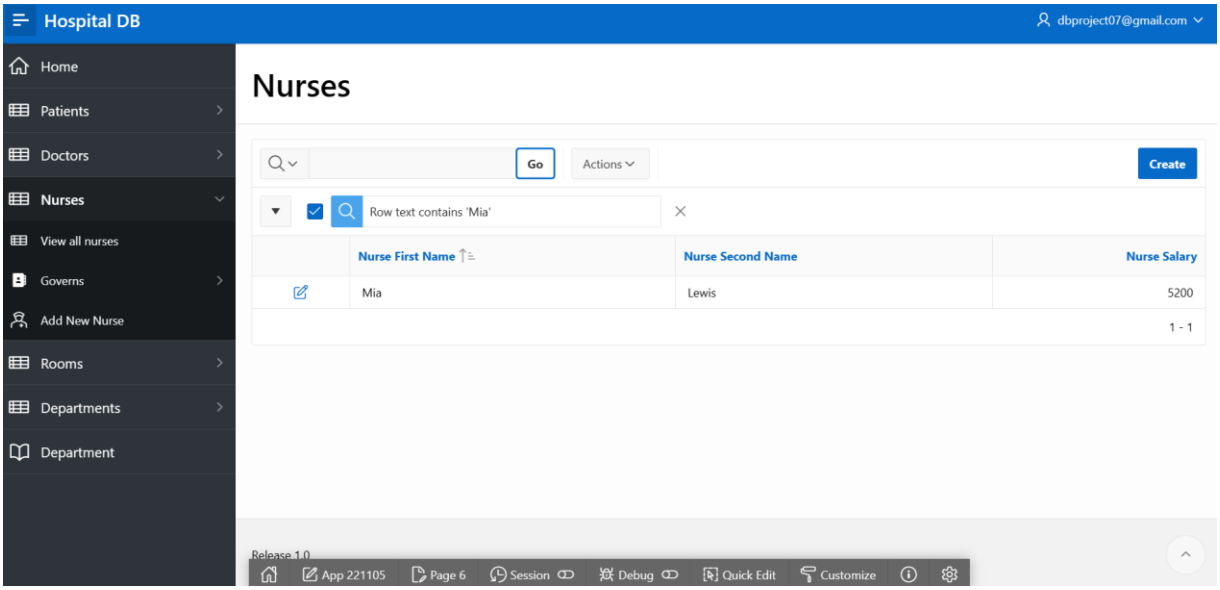


6- Nurse can update patient room number



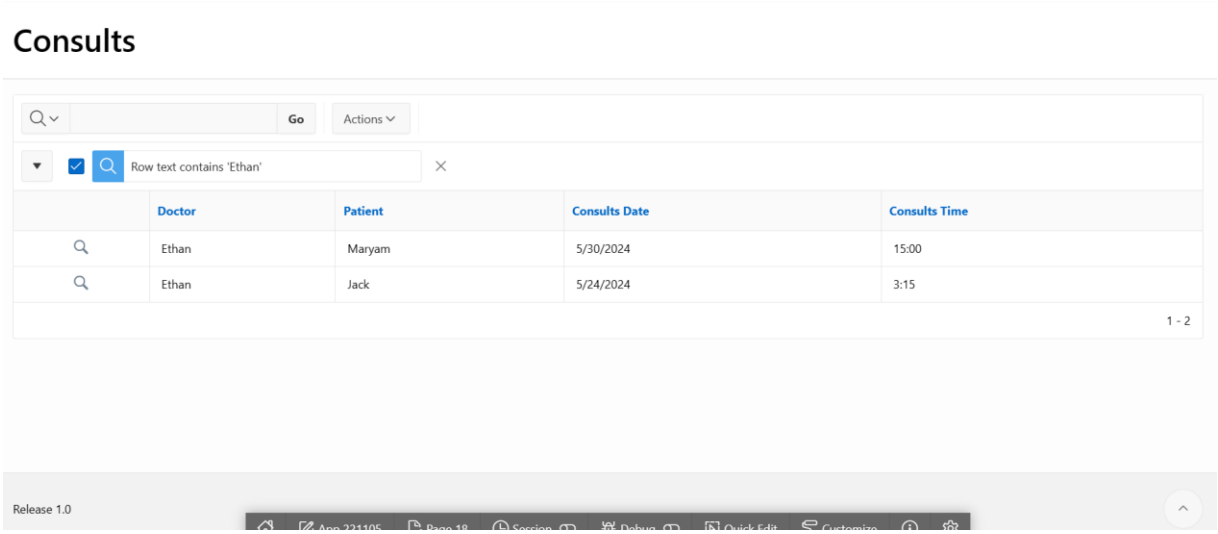
7-Doctor and nurse can view their salary record.



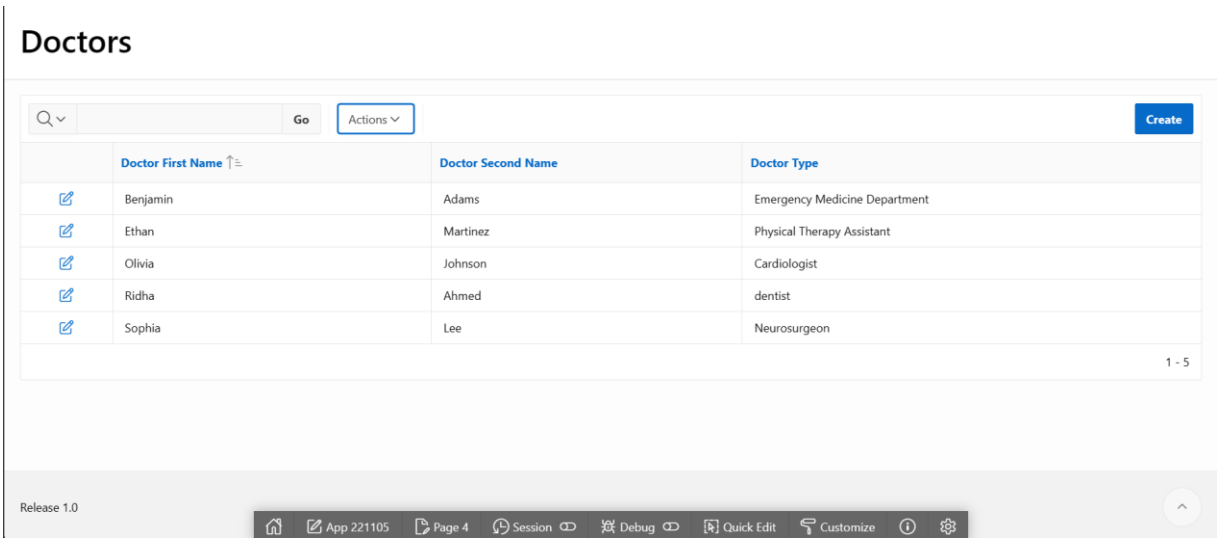


- Implement all reports which you have listed under 2-C, where the user should able to select few parameters and the system should provide a result summary.

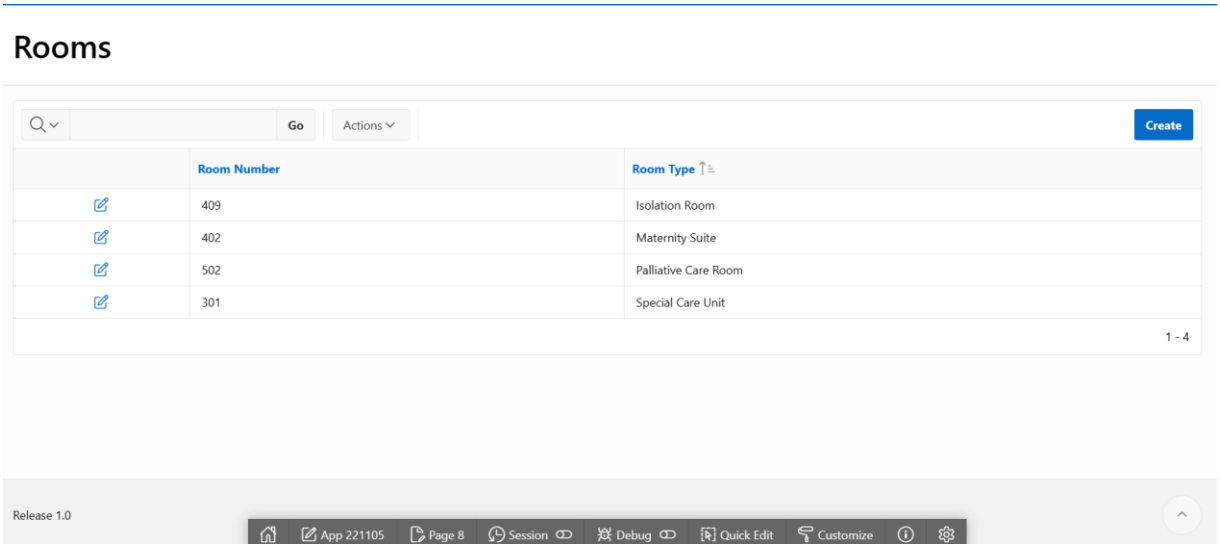
-Doctor can generate a list of expected patients.



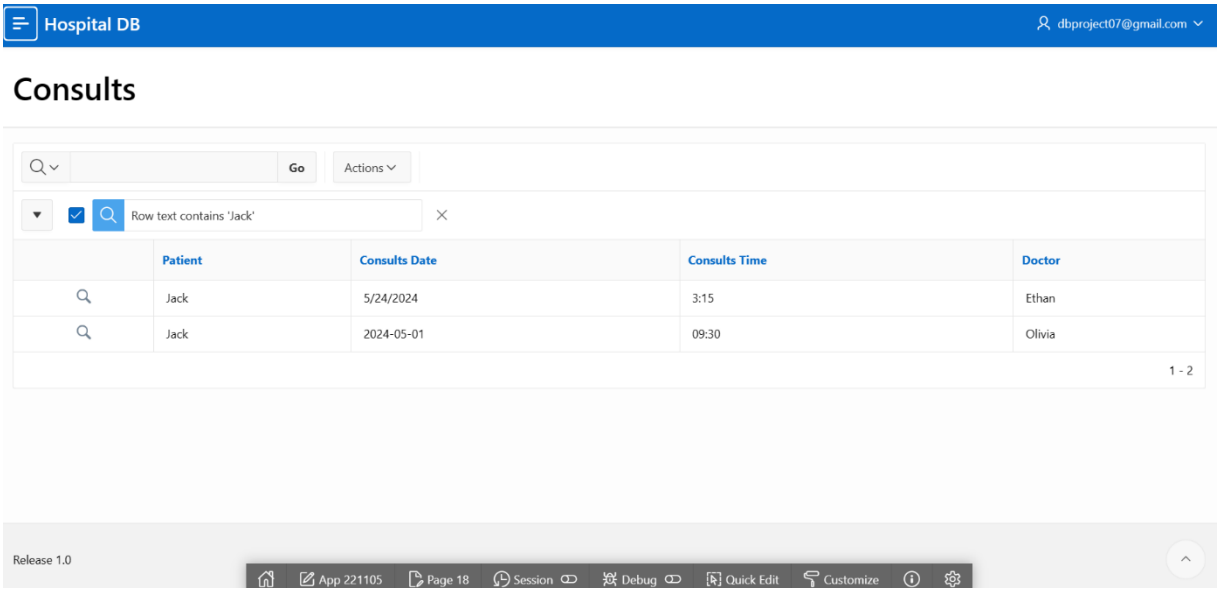
-Patient can generate a list of all doctors.



-Nurse can generate a list of all rooms.



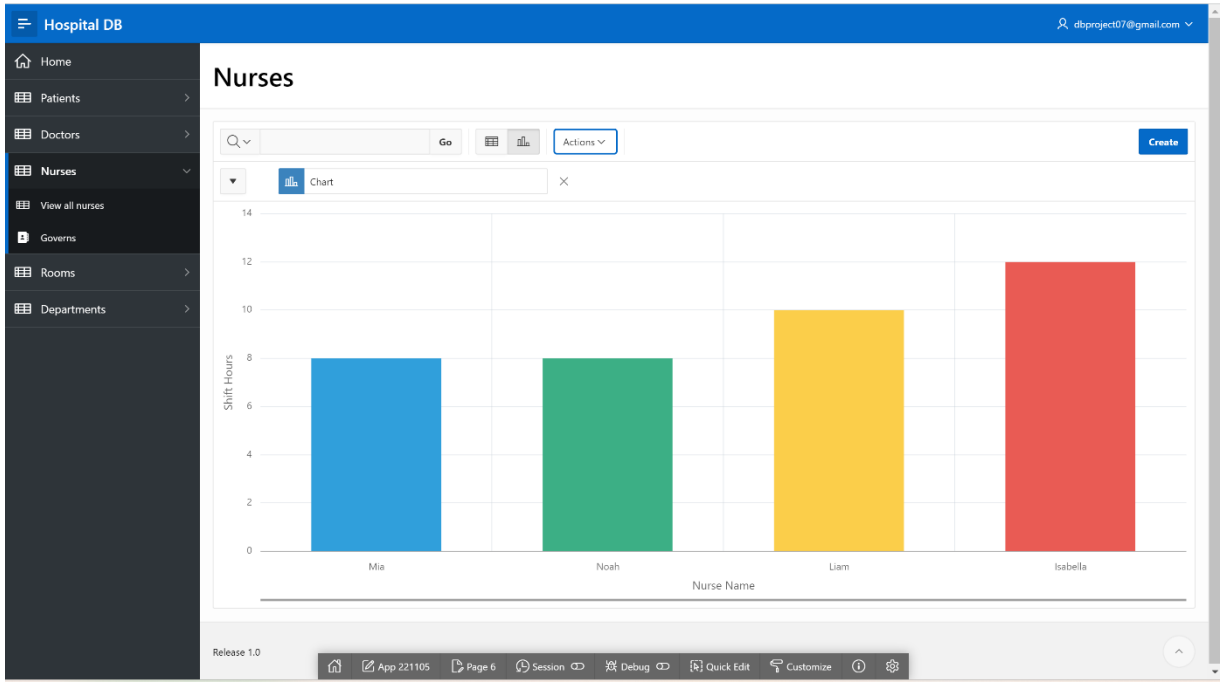
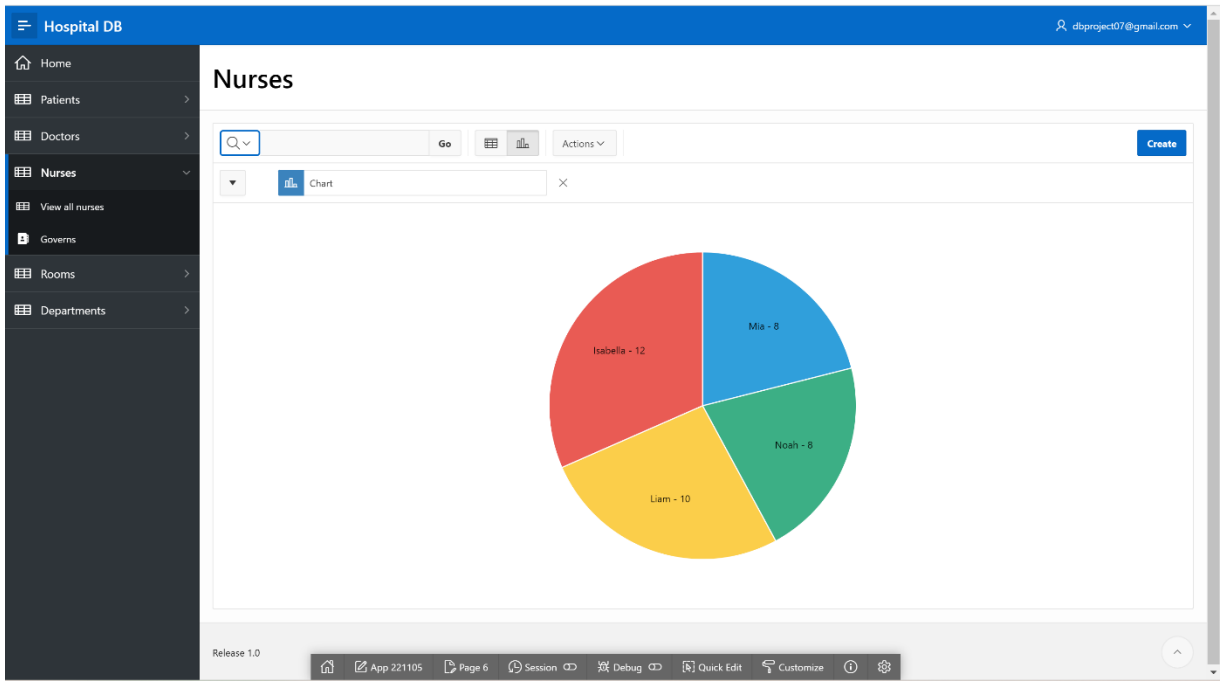
-Patient can generate a list of his/her upcoming appointments.



Extra:

Chart of statistics on shift hours for each nurse:

Bar chart:



Pie chart: